

Formula Student Rules 2027

Aerodynamics — shortened extract

English-only version. Includes T8 and provisions from other chapters that directly concern aerodynamic devices, their geometry, placement, structural requirements, vehicle clearance and post-inspection adjustment.

T 1 — Definitions relevant to aerodynamics

Original rules: pp. 18–19

T 1.1.2 Bodywork — the outermost surface of the chassis or any fairing parts and covers.

T 1.1.18 Surface envelope — the surface of the union of the rollover protection envelope, see T 1.1.16, and the volume defined by:

- the top of the main hoop and the outside edges of the four tires
- a plane from the top of the main hoop to the top of the front bulkhead
- a plane from the top of the main hoop to the rearmost structure according to T 3.2 or equivalent
- a projection in side view of the lower boundaries of the rollover protection envelope

Example depicted in Figure 3.

T 2 — General design requirements affecting aerodynamics

Original rules: pp. 20–22

T 2.1.2 The vehicle must be open-wheeled, single seat and open cockpit (a formula style body) with four wheels that are not in a straight line.

T 2.1.3 Open wheel vehicles must satisfy the following:

- The wheel/tire assembly must be unobstructed when viewed from the side.
- No part of the vehicle may enter a keep-out-zone defined by two lines extending vertically from positions 75 mm in front of and 75 mm behind the outer diameter of the front and rear tires in the side view of the vehicle, with steering system straight ahead. This keep-out zone extends laterally outward from the inboard plane of the wheel/tire assembly indefinitely. The keep-out-zone at the rear tires extends up to 700 mm above the ground.
- An additional keep-out-zone extends from the top of the rear tire to 700 mm above the ground and 150 mm inboard from the inboard plane of the rear tire. It is bounded longitudinally by the 75 mm keep-out-zone of the rear tire.

T 2.1.4 No part of the vehicle forward of the vertical plane through the leading edge of the front tires must enter the variable keep-out zone, see Figure 5. It is defined by two independent, non-intersecting, cuboids of 75 mm × 250 mm × infinite length. One infinite-length face must rest on the ground, and the infinite-length edges must be parallel to the vehicle longitudinal axis. The cuboids must not extend outboard of the front tires.

T 2.2.1 The minimum static ground clearance of any portion of the vehicle, other than the tires, including a driver, must be 30 mm. If an active suspension system is installed, the static ground clearance is measured in the lowest adjustable position.

T 2.2.2 Sliding skirts or other aerodynamic devices that by design, fabrication or as a consequence of moving, contact the track surface are prohibited.

T 2.3.2 In any side section view in front of the cockpit opening and outside the area defined in T 8.2 all parts of the bodywork must have no external concave radii of curvatures. Any gaps between bodywork and other parts must be reduced to a minimum.

T 2.3.4 The bodywork in front of the front wheels with the tangent of surface above 45° relative to the forward direction must have a radius of at least 38 mm along the top, sides and bottom of all affected edges.

T 2.4.1 For all edges that could come into contact with any standing pedestrian without reaching to the vehicle, the minimum radius of the bodywork and aerodynamic devices is 3 mm for all forward facing edges and 1 mm for all other edges.

T 3 — Chassis rules directly affecting aerodynamic devices

Original rules: pp. 33–35

T 3.19.4 Vehicles with aerodynamic devices and/or sensors forward of the front bulkhead must not exceed the peak deceleration of T 3.19.1 for the combination of their IA assembly and the non-crushable object(s). One of the following three methods must be used to prove the design does not exceed 120 kN:

- Physical testing of the IA assembly including any non-crushable object(s) or geometrically representative dummies thereof forward of the AIP. When non-crushable object(s) are located in front view of the AIP, physical testing is required.
- Combining the peak force from physical testing of the IA assembly with the failure load for the mounting of the non-crushable object(s), calculated from fastener shear and/or link buckling.
- Combining the “standard” IA peak load of 95 kN with the failure load for the mounting of the non-crushable object(s), calculated from fastener shear and/or link buckling.

T 3.20.1 All non-crushable objects (e.g. pedals, master cylinders, hydraulic reservoirs) must be rearward of the rear most plane of the front bulkhead and at least 25 mm behind the AIP at any time, except for sensors, aerodynamic devices and their mountings.

T 3.20.2 Sensor and aerodynamic devices may extend forward of the AIP, but their chassis attachment point must be located rearward of the AIP.

T 8 — Aerodynamic Devices

Original rule: p. 44

T 8.1.1 A specifically designed structure mounted on the vehicle to guide the airflow around the vehicle, increasing the downforce acting on the vehicle and/or lowering its drag. The mounting of this structure is not regarded as an aerodynamic device, unless it is intentionally designed to be one.

T 8.2.1 Height restrictions:

- All aerodynamic devices forward of a vertical plane through the leading edge of the front tires must be lower than 350 mm from the ground.
- All aerodynamic devices rearward of a vertical plane through the rearmost portion of the front face of the driver head restraint support, excluding any padding, set to its most rearward position must be lower than 1100 mm from the ground.
- All other aerodynamic devices must be lower than the virtual horizontal plane defined by the topmost points of the two tires on the left or right side of the vehicle, whichever side is closer to the device.

T 8.2.2 Width restrictions:

- All aerodynamic devices lower than the topmost points of the tires and further rearward than a vertical plane through the leading edge of the front tire, must not be wider than a vertical plane touching the most outboard point of the front and rear wheel/tire.
- All aerodynamic devices between the top edge of the rear tires and 700 mm above the ground, must not be wider than 150 mm inboard of the most inboard point of each rear tire.
- All aerodynamic devices between 700 mm and 1100 mm above the ground, must not extend outboard of the most outboard point of the rear wheel/tire.

T 8.2.3 Length restrictions:

- All aerodynamic devices must not extend further rearward than 250 mm from the rearmost part of the rear tires.
- All aerodynamic devices must not extend further forward than 700 mm from the fronts of the front tires.

T 8.2.4 All restrictions must be fulfilled with the wheels pointing straight and with any suspension setup with or without a driver seated in the vehicle.

T 8.3.1 Any aerodynamic device must be able to withstand a force of 200 N distributed over a minimum surface of 225 cm² and not deflect more than 10 mm in the load carrying direction.

T 8.3.2 Any aerodynamic device must be able to withstand a force of 50 N applied in any direction at any point and not deflect more than 25 mm.

A 6 — Vehicle movement rule relevant to wings

Original rule: p. 15

A 6.6.8 Vehicles with wings are required to have two team members walking on either side of the front wing whenever the vehicle is being pushed.

IN 1 — Technical Inspection rule relevant to aero setup

Original rule: p. 87

IN 1.5.1 After Technical Inspection, the only modifications allowed to the vehicle are:

- Adjustment of belts, chains and clutches
- Adjustment of the brake bias
- Adjustment of the driver restraint system, head restraint, seat and pedal assembly
- Substitution of the head restraint or seat insert for different drivers
- Adjustment to engine operating parameters, e.g. fuel mixture and ignition timing
- Adjustment of mirrors
- Adjustment of the suspension where no part substitution, other than springs, sway bars and shims, is required
- Adjustment of tire pressure
- Adjustment of winglet angles, but not the position of the complete aerodynamic device in relation to the vehicle
- Replenishment of fluids
- Replacement of defective tires or brake pads. Replacement tires and brake pads must be identical in material/composition/size to those presented and approved at Technical Inspection.
- Changing wheels and tires for “wet” or “damp” conditions as allowed in D 3.2 and D 7.6
- Software calibration changes
- Recharging LV batteries
- Recharging TS accumulators
- (De-)Coupling of actuators and sensor repositioning as allowed in T 14.7

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